

Sparse State Estimation under Sensor Corruption

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System Model

Discrete-time linear system

$$x_{k+1} = Ax_k + Bu_k + w_k \quad (1)$$

$$y_k = Cx_k + e_k + v_k \quad (2)$$

Variables

- ▶ $x_k \in \mathbb{R}^n$: state
- ▶ $u_k \in \mathbb{R}^m$: known input, w_k, v_k : Gaussian noises
- ▶ $y_k \in \mathbb{R}^p$: measurements
- ▶ $e_k \in \mathbb{R}^p$: sensor corruption

Assumption

Only a small number of sensors are corrupted (enforced using L_0 norm):

$$\|e_k\|_0 \leq s \ll p$$

Our goal: jointly estimate $\{x_k\}$ and identify corrupted sensors with their biases.

Formulation as an optimization problem (L_0 Formulation)

$$\begin{aligned} \min_{\{x_k, e_k\}} \quad & \sum_{k=0}^N \|y_k - Cx_k - e_k\|_2^2 \\ \text{subject to} \quad & \begin{cases} x_{k+1} = Ax_k + Bu_k \\ \|e_k\|_0 \leq s \end{cases} \end{aligned}$$

- ▶ e_k explains corrupted sensors
- ▶ L_0 constraint enforces sparsity

Caveat

The L_0 constraint requires combinatorial $\binom{p}{s}$ search over sensor subsets. This explodes even for moderate p .

Sparse Sensor Corruption

Stacking measurements over T steps gives

$$Y = \mathcal{O}_T x_0 + E,$$

where $\mathcal{O}_T$ is the observability matrix

$$\mathcal{O}_T = [C \quad CA \quad CA^2 \quad \dots \quad CA^{T-1}]^T$$

Proposition (Fawzi–Tabuada)

The initial state x_0 can be uniquely recovered despite s corrupted sensors iff for every nonzero x

$$|\{i : (Cx)_i \neq 0 \text{ or } (CAx)_i \neq 0 \dots (CA^{T-1}x)_i \neq 0\}| > 2s.$$

- ▶ The state must influence more than $2s$ sensor measurements across the observation horizon.
- ▶ Otherwise corruption of s sensors can mask change in state.

L_1 Convex Relaxation

We relax the L_0 constraint with an L_1 -Lasso type regularizer term.

$$\min_{\{x_k, e_k\}} \sum_{k=0}^N \|y_k - Cx_k - e_k\|_2^2 + \lambda \|e_k\|_1$$

subject to

$$x_{k+1} = Ax_k + Bu_k$$

Properties:

- ▶ Convex problem (L_2 norm + L_1 norm cost, linear equality constraints)
- ▶ L_1 penalty as a substitute for L_0 constraints
- ▶ Regularization is required, otherwise e_k can be exactly chosen to make cost = 0
- ▶ Solvable with solvers used for Lasso regression

Interpretation: Large residuals are attributed to sensor corruption rather than biasing the state estimate.

Batch vs Moving Horizon Formulations

Batch formulation

$$\min_{x_{0:N}, e_{0:N}} \sum_{k=0}^N \|y_k - Cx_k - e_k\|^2 + \lambda \|e_k\|_1$$

subject to dynamics.

- ▶ uses all data simultaneously
- ▶ offline estimation

Moving Horizon/Online Formulation

$$\min_{x_{k-N:k}, e_{k-N:k}} \sum_{i=k-N}^k \|y_i - Cx_i - e_i\|^2 + \lambda \|e_i\|_1$$

- ▶ sliding time window ($k - N$ to k)
- ▶ enables online estimation and implementation

Nonlinear Systems

For nonlinear dynamics

$$x_{k+1} = f(x_k, u_k)$$

$$y_k = h(x_k) + e_k$$

construct a dynamics-constrained optimization problem:

$$\min_{x,e} \sum_k \|y_k - h(x_k) - e_k\|^2 + \lambda \|e_k\|_1$$

subject to

$$x_{k+1} = f(x_k, u_k)$$

Typically solved via

- ▶ Sequential convex programming (by linearizing the dynamics locally) (OSQP)
- ▶ Interior point methods (IPOPT)

This yields a nonlinear moving-horizon estimator.

Sequential Convex Programming for Nonlinear Estimation

Linearization around current iterate $\bar{x}_k$

$$f(x_k, u_k) \approx f(\bar{x}_k, u_k) + A_k(x_k - \bar{x}_k)$$

$$h(x_k) \approx h(\bar{x}_k) + H_k(x_k - \bar{x}_k)$$

where

$$A_k = \frac{\partial f}{\partial x}(\bar{x}_k, u_k), \quad H_k = \frac{\partial h}{\partial x}(\bar{x}_k)$$

Convex subproblem (QP)

$$\min_{x, e} \sum_k \|y_k - h(\bar{x}_k) - H_k(x_k - \bar{x}_k) - e_k\|_2^2 + \lambda \|e_k\|_1$$

subject to

$$x_{k+1} = f(\bar{x}_k, u_k) + A_k(x_k - \bar{x}_k)$$

This convex QP is solved iteratively until convergence (via OSQP).

Robust Min–Max Estimation with Bounded Corruption

Assume sensor corruption is bounded but otherwise arbitrary:

$$\|e_k\|_\infty \leq \rho$$

Robust min–max estimator

$$\min_{x_{0:N}} \max_{\|e_k\|_\infty \leq \rho} \sum_{k=0}^N \|y_k - h(x_k) - e_k\|_2^2$$

subject to dynamics

$$x_{k+1} = f(x_k, u_k)$$

- ▶ Inner max chooses the worst corruption within the bound
- ▶ Outer min finds state estimate that minimizes worst-case error
- ▶ Can be formulated as a convex semi-infinite program (CSIP) in many cases

This yields a robust estimator that is resilient to bounded adversarial disturbances.

Related Literature

- ▶ H. Fawzi, P. Tabuada, S. Diggavi
Secure Estimation and Control for Cyber-Physical Systems Under Adversarial Attacks
IEEE TRANSACTIONS ON AUTOMATIC CONTROL, 2014
- ▶ S. Mishra et al.
Secure State Estimation Against Sensor Attacks in the Presence of Noise
IEEE TRANSACTIONS ON CONTROL OF NETWORK SYSTEM, 2017